

Voice Command Shopping Assistant - Technical Approach

I built a voice-based shopping list manager using React and the Web Speech API. The application features voice command recognition, smart suggestions, and shopping list management.

For voice input, I used the Web Speech API for voice recognition across multiple languages (English, Spanish, French, German). For natural language processing, I implemented a custom parser using Compromise.js that handles varied phrases like "Add milk" or "I need apples" and extracts items, quantities, and actions.

For smart suggestions, I built a recommendation engine that provides personalized suggestions based on purchase history, seasonal items (summer: watermelon, strawberries), and substitutes (e.g., almond milk for regular milk). The system tracks purchase history in localStorage to identify frequently bought items and running-low alerts.

For shopping list management, items are automatically categorized (dairy, produce, snacks, etc.) with quantity management and real-time visual feedback. The interface is mobile-optimized with a minimalist design.

The application is hosted on Vercel with continuous deployment from GitHub. Voice recognition works across modern browsers using the native Web Speech API. The project demonstrates clean, production-quality code with proper error handling, loading states, and comprehensive documentation. All required features are implemented including voice input, smart suggestions, shopping list management, voice-activated search, and multilingual support.